

① M/M/3/4, $\lambda=15$, $\mu=10$

(M10) $i=0 (W_0), W_0=1$
 $W_i = W_{i-1} \left(\frac{\lambda}{\mu}\right)$, mas com $C>1$ (3): $W_i = W_{i-1} \left(\frac{\lambda}{\min(i,C) \cdot \mu}\right)$

$$\begin{aligned} W_0 &= 1 \\ W_1 &= 1 \cdot \frac{15}{10} = 1,5 \\ W_2 &= 1,5 \cdot \frac{15}{20} = 1,125 \\ W_3 &= 1,125 \cdot \frac{15}{30} = 0,5625 \\ W_4 &= 0,5625 \cdot \frac{15}{30} = 0,28125 \end{aligned}$$

$$\pi_i = \frac{W_i}{\sum_{j=0}^4 W_j} \rightarrow \pi_4 = \frac{0,28125}{1+1,5+1,125+0,5625+0,28125}$$

$$\pi_4 = \frac{0,28125}{4,46875} = 0,0629370629$$

5

6,2937%

Perda = taxa de chegada ~~per~~ pela probabilidade de estar no estado 4. (cheio, último)

$15 \cdot 0,0629370629 \rightarrow$ ~~0,94405~~ clientes por hora.

0,94405

② $T_R = W = \frac{N}{D}$

$Var_{\pi}(0) = \sum_{j=1}^K (\pi_j \cdot \mu_j)$

Pop. Média (N) = $\sum_{j=1}^K (\pi_j \cdot j)$

$\pi_0 = \frac{1}{4,46875} = 0,223776$

$\pi_1 = \frac{1,5}{5} = 0,335664$

$\pi_2 = \frac{1,125}{5} = 0,251748$

$\pi_3 = \frac{0,5625}{5} = 0,125874$

$x \cdot j$	
	$\pi_0 = 0,223776$
x1	$0,335664$
x2	$0,503496$
x3	$0,377622$
MONTJ x4	$0,251748$
N=	$1,468531$

$\pi_1 \cdot 10 = 3,356643$

$\pi_2 \cdot 20 = 5,034965$

$\pi_3 \cdot 30 = 3,776223$

$\pi_4 \cdot 30 = 1,881111$

$D = 14,055944$

$W = \frac{N}{D} = \frac{1,468531}{14,055944} = 0,104477$
 horas

Tempo de resposta (W) por minuto $\approx 6,26862$